

ROYAUME DU MAROC • ENSEIGNEMENT SECONDAIRE QUALIFIANT

Pedagogical Lesson Plans

Physics & Chemistry (Common Core of Science & Technology)

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Approved Textbooks – Pedagogical Orientations – Ministerial Notes 09-142 & 144

Level: Common Core	Subject: Physics-Chemistry • Physics Component	Pedagogical Lesson Plans
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Part 1: Mechanics (Total: 28 Hours) • Module 1: Fundamental Interactions (6 Hours)

Targeted Specific Competencies: Exploit mechanical models for equilibrium of static/moving bodies. Raise awareness of road safety and speed hazards. Utilize the scientific method and simulation software. Connect everyday phenomena to physics theories.

Part Units Breakdown: 1. Universal Gravitation (4h) • 2. Mechanical Actions (2h) • 3. Motion (6h) • 4. Inertia Principle (4h) • 5. 2-Force Equilibrium (3h) • 6. 3-Force Equilibrium (4h) • 7. Rotation Equilibrium (5h)

Unit 1: Universal Gravitation		4 Hours	
Prerequisites: Mechanical actions; contact vs distance actions; weight and mass; action-reaction principle.		Problem Situation: The solar system maintains planets in orbit. What ensures this cosmic cohesion? What distinguishes weight from universal gravitational attraction?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Scale of distances & orders of magnitude. - Universal gravitation law: $F = G(m_A \cdot m_B)/d^2$. - Terrestrial gravity g and weight $P = m \cdot g$. - Variation of g with altitude.	I. Distance Scale (1. Powers of 10, 2. Scale axis) II. Gravitation (1. Newton's law, 2. Application) III. Gravity & Weight (1. Body weight, 2. Variations of g)	- Textbook, board, computer media - Astronomical videos & solar system charts - Spring dynamometers, slotted masses	Activities: Teacher guides planetary force calculations. Students calculate gravitational attraction and deduce order of magnitude. Evaluation: Diagnostic Qs, formative exercises, Supervised Exam 1.

Unit 2: Examples of Mechanical Actions		2 Hours	
Prerequisites: Mechanical actions, force vector characteristics, measurement and representation.		Problem Situation: Water exerts immense forces on dam walls. What are the classifications of these forces and how is pressure defined?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Inventory of forces (contact vs remote). - Localized vs distributed contact forces. - Pressing force & pressure formula $P = F / S$. - Pressure units (Pa, bar, hPa).	I. Force Concept (1. Definition, 2. Force vector) II. Classification (1. Contact forces, 2. Distance forces, 3. Internal/External) III. Pressing Force & Pressure (1. $P = F/S$, 2. Pressure units)	- Springs, wooden blocks, balloons - Syringe and digital manometer - Sand tray, weights, air pump	Activities: Teacher demonstrates pressure variations on sand. Students perform force classification and calculate pressure. Evaluation: Formative problem sets and pressure unit conversions.

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Part 1: Mechanics • Module 2: Motion and Inertia Principle (10 Hours)

Unit 3: Motion		6 Hours	
Prerequisites: Average speed; road safety; uniform/accelerated motion; translation vs rotation.		Problem Situation: An escalator rider is at rest relative to the stairs but in motion relative to Earth. What defines relativity of motion? How is velocity represented?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Reference frame (space and time frames). - Trajectory and types of motion. - Average and instantaneous speed. - Instantaneous velocity vector representation. - Uniform rectilinear motion equation: $x(t) = v \cdot t + x_0$. - Uniform circular motion (period, frequency).	I. Relativity of Motion (1. Reference body, 2. Space/time frame, 3. Trajectory) II. Velocity (1. Average speed, 2. Instantaneous velocity vector) III. Uniform Rectilinear Motion (1. Characteristics, 2. Time equation) IV. Uniform Circular Motion	- Air cushion table, pucks, spark timer - Avimeca motion tracking software, webcam - Tracing paper, metric rulers, chronometers	Activities: Analyze spark records to compute $v_i = M_{(i-1)}M_{(i+1)} / (2\tau)$. Draw velocity vectors and write $x(t)$ equations. Evaluation: Trajectory analysis exercises and time equation problems.

Unit 4: The Principle of Inertia		4 Hours	
Prerequisites: Forces, velocity vector, trajectory, reference frames.		Problem Situation: Bus passengers jerk forward when brakes are applied. Is a force necessary to maintain motion? What is the center of inertia?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Identify the center of inertia G of a solid. - Define mechanically isolated and semi-isolated systems. - State and apply the Principle of Inertia (Newton's 1st Law). - Apply the barycentric center of mass formula.	I. Center of Inertia G (1. Definition, 2. Motion of G vs other points) II. Principle of Inertia (1. Semi-isolated system, 2. 1st Law statement: $\Sigma F_{ext} = 0 \Leftrightarrow v_G = \text{const}$) III. Center of Mass (1. Barycentric relation, 2. Compound systems)	- Air cushion table with self-supporting pucks - Puck with eccentric mass / coupled pucks - Spark recording paper and rulers	Activities: Record motion of asymmetric glider; verify that G follows uniform rectilinear motion while the body rotates. Evaluation: Problem sets on center of mass and isolated systems.

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Part 1: Mechanics • Module 3: Equilibrium of Solid Bodies (12 Hours)					
Unit 5: Equilibrium of a Solid Subject to Two Forces					3 Hours
Prerequisites: Forces, vectors, body weight, action-reaction.			Problem Situation: Automobile springs ensure ride comfort and balance. What characterizes a spring? Why do ships float on water?		
Objectives		Contents	Didactic Materials		Activities & Evaluation
- Equilibrium condition under 2 forces: $F_1 + F_2 = 0$. - Hooke's spring tension law: $T = k \cdot \Delta l$. - Archimedes' upthrust: $F_A = \rho_{\text{fluid}} \cdot V \cdot g$. - Static friction coefficient and friction angle φ.		I. 2-Force Equilibrium (1. Equilibrium condition) II. Spring Force (1. Hooke's Law, 2. Stiffness k) III. Fluid Thrust & Friction (1. Archimedes' law, 2. Friction on plane)	- Helical springs of varied stiffness, slotted masses - Dynamometers, test tube, water, oil - Inclined plane apparatus with wooden block		Activities: Plot $T = f(\Delta l)$ to determine stiffness k . Measure apparent weight loss in liquid to verify Archimedes' principle. Evaluation: Spring calibration exercises and floating bodies problems.
Unit 6: Equilibrium under Three Non-Parallel Forces					4 Hours
Prerequisites: 2-force equilibrium, vector addition, orthogonal projection.			Problem Situation: A rock climber suspended by ropes stays in equilibrium. What mathematical conditions ensure equilibrium under 3 forces?		
Objectives		Contents	Didactic Materials		Activities & Evaluation
- Coplanarity and concurrency conditions. - Vector equilibrium condition: $F_1 + F_2 + F_3 = 0$. - Geometric resolution: closed vector polygon. - Analytical resolution: projection on Ox and Oy.		I. Experimental Study (1. Concurrency, 2. Coplanarity) II. Equilibrium Conditions (1. Vector sum, 2. Closed polygon) III. Analytical Method (1. Projection on axes, 2. Resolution)	- Cardboard plate with negligible mass 3 dynamometers with hooks, white sheet, protractor - Magnetic whiteboard pins and pulleys		Activities: Assemble 3-force balance on plate. Construct the vector triangle and verify closure. Project forces along Cartesian axes. Evaluation: Equilibrium problem solving and Supervised Exam 3.

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Part 1: Mechanics • Module 3: Equilibrium (Continued)

Unit 7: Equilibrium of a Solid Capable of Rotating Around a Fixed Axis			5 Hours
Prerequisites: Forces, 2-force and 3-force equilibrium, rotation motion.		Problem Situation: When changing a flat tire, drivers use a lug wrench. Why does a longer wrench loosen the bolt more easily? What physical quantity measures rotational effect?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Understand rotational effect of a force. - Moment formula: $M_{\Delta}(F) = \pm F \cdot d$ (N·m). - Theorem of Moments: $\sum M_{\Delta}(F) = 0$. - Couple of two forces: $M_{\Delta} = F \cdot d$. - Torsion couple and wire constant C: $M_c = -C \cdot \theta$.	I. Moment of a Force (1. Rotational effect, 2. Definition & formula $M_{\Delta}(F) = \pm F \cdot d$) II. Equilibrium in Rotation (1. Experimental study on disc, 2. Theorem of moments: $\sum M = 0$, 3. Full equilibrium conditions) III. Couples of Forces (1. Couple of two forces, 2. Torsion couple $M_c = -C \cdot \theta$, 3. Torsion constant C)	- Equilibrium disc with ball bearings, perforated bar - Dynamometers, pulleys, weights, threads - Torsion pendulum apparatus with protractor scale	Activities: Vary force distance on equilibrium disc; verify that $F_1 \cdot d_1 = F_2 \cdot d_2$. Measure wire torsion angle θ under opposing torque to calculate C. Evaluation: Moment calculation exercises, crane stability problems, Exam 3.

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References: Approved Textbooks – Pedagogical Orientations – Ministerial Memos 09-142 & 144

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Semester 2: Electricity (Total: 32 Hours) • Module 1: Current and Voltage (6 Hours)				
Unit 1: Direct Electric Current				3 Hours
Prerequisites: Atomic structure, electrons, conductors and insulators, multimeters.		Problem Situation: Lightning delivers 100 to 200 kA in a millisecond. What is electric current? How is current intensity defined, measured, and distributed in circuits?		
Objectives	Contents	Didactic Materials	Activities & Evaluation	
- Nature of electric current (electrons vs ions). - Conventional direction (+ to -) & quantity $Q = I \cdot t = n \cdot e$. - Current intensity in Amperes (A) & ammeter usage. - Kirchhoff's Node Law in series and parallel loops.	I. Types of Electricity (1. Friction charging, 2. Positive/negative) II. Direct Current (1. Nature in metals & electrolytes, 2. Conventional direction, 3. Intensity definition) III. Current Laws (1. Series circuit, 2. Node law in branched circuit: $\Sigma I_{in} = \Sigma I_{out}$)	- DC power supplies, lamps, switches, ammeters - Ebonite/glass rods, cloth, electroscope - Electrolysis cell with NaCl solution	Activities: Connect ammeters in series; verify Node Law at junctions. Evaluation: Node law exercises and ammeter reading calculations.	
Unit 2: Electric Voltage				3 Hours
Prerequisites: Current, circuit loops, voltmeter usage.		Problem Situation: Electrocardiograms display cardiac pulses as variable voltages. What is voltage? How is AC sinusoidal voltage characterized?		
Objectives	Contents	Didactic Materials	Activities & Evaluation	
- Voltage $U_{AB} = V_A - V_B$ & arrow representation. - Algebraic property: $U_{AB} = -U_{BA}$. - Loop rule (addition of voltages in loops). - Oscilloscope readings: U_{max}, $U_{eff} = U_{max}/\sqrt{2}$, T, f.	I. Electric Voltage (1. Definition, 2. Voltmeter measurement, 3. Representation) II. Potential & Loop Rule (1. Potential difference, 2. Addition of voltages) III. Alternating Voltages (1. Sinusoidal voltage, 2. Oscilloscope display: U_{max} and T)	- Low-frequency generator (GBF), oscilloscope - Digital and analog needle voltmeters - Resistors, LEDs, AC/DC power sources	Activities: Connect voltmeter in parallel; apply Loop Rule in multi-branch circuits. Measure peak voltage and period on oscilloscope. Evaluation: Oscilloscope reading tests and loop calculations.	

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Semester 2: Electricity • Module 2: Circuit Assemblies (13 Hours)

Unit 3: Resistor Associations 3 Hours

Prerequisites:

Current, voltage, Ohm's law, multimeters.

Problem Situation:

Motherboards contain hundreds of resistors in series and parallel. How do resistors divide voltage, and why is copper preferred for wires?

Objectives	Contents	Didactic Materials	Activities & Evaluation
- Ohm's law: $U = R \cdot I$; conductance $G = 1/R$ (S). - Equivalent resistance in series: $R_{eq} = \sum R_i$. - Equivalent resistance in parallel: $1/R_{eq} = \sum 1/R_i$. - Conductor resistance formula: $R = \rho \cdot l / S$. - Voltage divider circuit formula: $U_2 = E \cdot R_2 / (R_1 + R_2)$.	I. Ohmic Resistor (1. Definition & Ohm's law, 2. Wire resistivity $R = \rho \cdot l / S$) II. Resistor Combinations (1. Series combination, 2. Parallel combination) III. Applications (1. Voltage divider circuit, 2. Rheostat)	- Multimeters, DC power supplies (6V) - Resistor kits of various ohmic values - Wires of identical cross-section but different metals (Cu, Fe, Ni-Cr)	Activities: Measure equivalent resistance with ohmmeter; assemble a voltage divider and verify theoretical formula. Evaluation: Equivalent resistance calculations and voltage divider tests.

Unit 4: Passive Non-Linear Dipoles 5 Hours

Prerequisites:

Current, voltage, Ohm's law, multimeters.

Problem Situation:

Smartphones and electronic devices use diodes, LEDs, and thermistors. What are passive dipoles, what are their I-V curves, and where are they used?

Objectives	Contents	Didactic Materials	Activities & Evaluation
- Distinguish active vs passive dipoles. - Plot I-V characteristic curve of dipoles. - Study junction diode (threshold voltage $U_s \approx 0.6V$). - Study Zener diode (Zener voltage U_Z for voltage regulation). - Understand sensors: Thermistors (CTN/CTP) and LDR (photoresistor).	I. Dipole Concept (1. Definition, 2. Passive dipoles) II. Characteristic Curve (1. Test circuit, 2. Curve plotting) III. Various Passive Dipoles (1. Incandescent lamp, 2. Semiconductor diode, 3. Zener diode, 4. LED, 5. VDR varistor, 6. Thermistor, 7. LDR)	- Variable DC power supply (0 - 15 V), breadboard - Diodes (1N4007), Zener diodes, LEDs, thermistors, LDR - Incandescent lamp, torchlight, hair dryer (for heat)	Activities: Record (U, I) points in forward and reverse bias; plot curves on millimeter paper and deduce threshold voltage. Evaluation: Questions on diode rectification and sensor resistance shifts.

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Semester 2: Electricity • Module 2 & 3: Active Dipoles and Transistors (14 Hours)

Unit 5: Active Dipoles			4 Hours
Prerequisites: DC current, Ohm's law, multimeters, passive dipoles.		Problem Situation: Batteries act as generators while electric motors act as receivers. What physical equations govern their performance, and how is the operating point determined?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Linear generator equation: $U_{PN} = E - r \cdot I$. - Receiver equation: $U_{AB} = E' + r' \cdot I$. - Determine operating point (I_P, U_P) graphically. - Apply Pouillet's Law: $I = (\Sigma E - \Sigma E') / (\Sigma R)$. - Series/parallel associations of generators.	I. Active Dipole (1. Definition, 2. Generator characteristic, 3. EMF E and internal resistance r) II. Electric Receiver (1. Definition, 2. Back-EMF E' and r') III. Operating Point (1. Generator-resistor circuit, 2. Pouillet's Law)	- Flat 4.5V batteries, DC power supply, resistors - Electrolyzer with copper chloride solution - Rheostat, switch, multimeters, millimeter paper	Activities: Plot load line of battery and resistance line on same graph; find intersection point (I_P , U_P); compare with Pouillet's law. Evaluation: Pouillet's law problems and generator coupling calculations.

Unit 6: Simple Electronic Circuits - The Transistor			10 Hours
Prerequisites: Current, voltage, Ohm's law, diode characteristics.		Problem Situation: Modern electronics relies on transistors as microscopic switches and amplifiers. How does an NPN transistor control a large output current with a minute base signal?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Identify terminals of NPN transistor (B, C, E). - Current relationship: $I_E = I_B + I_C$. - Linear amplification: $I_C = \beta \cdot I_B$. - Switching modes: Blocked ($I_B = 0$, $I_C = 0$) vs Saturated. - Assemble sensor circuits (light-activated switch with LDR).	I. Bipolar Transistor (1. Terminals & symbol, 2. Current relations $I_E = I_B + I_C$) II. Operating Regimes (1. Linear amplification regime $I_C = \beta \cdot I_B$, 2. Saturation and cutoff regimes) III. Practical Assemblies (1. Twilight automatic switch, 2. Temperature alarm)	- NPN transistors (2N2222, BC548), breadboard - Resistors (1k, 10k, 100k), LEDs, buzzer, relays - LDR photoresistors, thermistors (CTN), multimeters	Activities: Wire transistor circuit; measure I_B and I_C to compute current gain β . Build twilight alarm and test with covered LDR. Evaluation: Transistor state analysis problems and Supervised Exam 6.

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Chemistry Component • Semester 1: Chemistry Around Us (Total: 10 Hours)

Competencies: Relate everyday products to chemistry principles. Safe use of lab equipment. Chemical synthesis protocols following safety rules. Experimental methods.

Semester Units: 1. Chemical Species (2h) • 2. Extraction & Separation (3h) • 3. Chemical Synthesis (3h) • 4. Atomic Model (2h) • 5. Molecular Geometry (2h) • 6. Periodic Classification (4h)

Unit 1: Chemical Species		2 Hours	
Prerequisites: States of matter, mixtures vs pure substances, combustion.		Problem Situation: Natural products and manufactured goods contain myriad substances. How do chemists classify chemical species and test for water, glucose, and starch?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Define chemical species and chemical substance. - Classify species into natural and synthetic. - Identification test for water (anhydrous CuSO_4 turns blue). - Identification test for glucose (Fehling test: red precipitate). - Identification test for acidity (pH paper / indicators).	I. Concept of Chemical Species (1. Definition, 2. Chemical tests) II. Classification (1. Natural chemical species, 2. Synthetic chemical species, 3. Artificial species)	- Anhydrous copper sulfate, Fehling solution A & B - Orange fruit juice, apple slice, potato, salt - Test tubes, test tube rack, Bunsen burner, pH paper	Activities: Perform tests for water in fruits with white CuSO_4 . Perform Fehling test on fruit juices to detect reducing sugars. Evaluation: Diagnostic quiz on tests and natural/synthetic identification.

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Semester 1: Chemistry • Module 1: Extraction, Separation and Synthesis (6 Hours)

Unit 2: Extraction, Separation and Detection of Chemical Species		3 Hours	
Prerequisites: Mixtures, solubility, boiling point, density.		Problem Situation: Brewing tea extracts flavors and colors using hot water as solvent. What are extraction techniques, and how does Thin Layer Chromatography separate components?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Know extraction techniques (solvent extraction, hydrodistillation). - Use a separatory funnel properly based on density. - Thin-Layer Chromatography (TLC): stationary & mobile phase. - Calculate frontal ratio: $R_f = h / H$.	I. Extraction Techniques (1. Solvent extraction, 2. Hydrodistillation setup) II. Separation (1. Liquid-liquid extraction, 2. Separatory funnel usage) III. Chromatography (TLC) (1. Principle, 2. Development & visualization, 3. Frontal ratio R_f)	- Hydrodistillation glassware: round flask, condenser - Lavender flowers, cyclohexane, dichloromethane - TLC silica plates, developing tank, UV lamp, capillary tubes	Activities: Perform hydrodistillation of lavender; separate organic phase in separatory funnel; run TLC and measure R_f . Evaluation: TLC chromatogram reading and extraction protocol tests.

Unit 3: Synthesis of Chemical Species		3 Hours	
Prerequisites: Chemical species, extraction, glassware, reflux heating.		Problem Situation: Industrial chemistry produces fragrances identical to natural ones. Why synthesize species instead of extracting them, and how is reflux heating carried out?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Understand motives for chemical synthesis (cost, purity, supply). - Assemble reflux heating apparatus safely. - Synthesize linalyl acetate (lavender scent). - Compare synthetic product to natural extract via TLC.	I. Necessity of Synthesis (1. Cost and availability, 2. Environmental conservation) II. Reflux Synthesis (1. Principle of reflux without vapor loss, 2. Synthesis protocol) III. Characterization (1. Washing & drying, 2. TLC comparison)	- Reflux setup: heating mantle, Liebig condenser, boiling stones - Linalool, acetic anhydride, sodium hydrogencarbonate - Separatory funnels, beakers, TLC equipment	Activities: Heat linalool and acetic anhydride under reflux for 20 min. Wash organic phase and spot TLC against natural lavender oil. Evaluation: Questions on reflux role, washing steps, and TLC analysis.

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Semester 1: Chemistry • Module 2: The Atom and Molecular Geometry (12 Hours)

Unit 4: The Atomic Model		2 Hours	
Prerequisites: Atom components, electrons, ions, electrical conductivity.		Problem Situation: Rutherford discovered the atomic nucleus in 1911. What is the modern model of the atom, how are electrons distributed in shells, and what are isotopes?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Constituents of atom: nucleus (protons Z, neutrons N) and electrons. - Nuclide notation ${}^A_Z X$ and atomic neutrality. - Isotopes definition and atomic mass concentrated in nucleus. - Electronic structure in shells (K)², (L)⁸, (M)⁸.	I. Atomic Structure (1. Nucleus & nucleons, 2. Electrons, 3. Atomic mass) II. Chemical Element (1. Atomic number Z, 2. Isotopes, 3. Monatomic ions) III. Electron Distribution (1. Shells K, L, M, 2. Electronic configuration)	- Textbook, whiteboard, atomic simulations - Posters of Rutherford and Bohr models - Chemical reagents: copper turnings, nitric acid (conservation of Cu)	Activities: Demonstrate conservation of copper element through cycle of reactions. Write electron configurations for elements Z = 1 to 18. Evaluation: Isotope counting and electronic configuration quizzes.

Unit 5: Molecular Geometry		2 Hours	
Prerequisites: Electronic structure, valence electrons, ions.		Problem Situation: Atoms bind together to form molecules with specific 3D geometries. What rules govern chemical bonding, and how do we draw Lewis and Cram representations?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Duet (2) and Octet (8) stability rules. - Covalent bonding: bonding and non-bonding electron pairs. - Lewis structures of molecules (H₂O, CH₄, NH₃, CO₂). - Isomerism: structural isomers. 3D spatial geometry: Cram notation (tetrahedral, pyramid, bent, linear).	I. Duet & Octet Rules (1. Inert gas configurations, 2. Monatomic ions stability) II. Lewis Model (1. Covalent bond, 2. Lewis representation, 3. Isomers) III. Molecular Geometry (1. Electron pair repulsion, 2. Cram 3D representation)	- Molecular model kits (plastic balls and connectors) - Computer software for 3D molecular visualization (ChemSketch) - Whiteboard and colored markers	Activities: Build physical models of methane CH ₄ , ammonia NH ₃ , and water H ₂ O; deduce valence bond angles and draw Cram formulas. Evaluation: Lewis diagram worksheets and isomer identification tests.

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Semester 2: Chemistry • Transformations of Matter (Total: 22 Hours)

Unit 6: Periodic Classification of Elements		4 Hours	
Prerequisites: Atomic number Z, electronic configuration (K, L, M), valence electrons.		Problem Situation: Mendeleev organized elements into a periodic table in 1869. What criteria govern modern periodic classification, and what properties do elements in the same column share?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Understand historical construction (Mendeleev) based on atomic mass. - Modern criteria: ordered by increasing atomic number Z. - Rows (periods): same number of occupied shells. - Columns (chemical families): same valence electrons. - Identify alkali metals, alkaline earths, halogens, and noble gases.	I. Mendeleev's Classification (1. Historical context, 2. Predicted elements) II. Modern Periodic Table (1. Periods & columns, 2. Electronic configuration relationship) III. Chemical Families (1. Alkali metals (Col 1), 2. Halogens (Col 17), 3. Noble gases (Col 18))	- Large wall periodic table of chemical elements - Solutions: NaCl, NaBr, NaI, AgNO₃ (precipitation of halides) - Chlorine water, bromine water, iodine solution	Activities: Test similarities between halogens Cl ⁻ , Br ⁻ , I ⁻ using silver nitrate. Locate elements in the table from their electron formula. Evaluation: Periodic table mapping exercises and family identification.

Unit 7: Tools for Describing a Chemical System (The Mole)		4 Hours	
Prerequisites: Mass, volume, atoms, molecules, density.		Problem Situation: A single drop of water contains 10 ²¹ molecules. How do chemists count such astronomical numbers of particles, and what is the mole?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Define the mole and Avogadro constant: N_A = 6.022 × 10²³ mol⁻¹. - Molar mass M: atomic, molecular, and ionic (g/mol). - Relation $n = m / M$ and $n = (\rho \cdot V) / M$. - Molar volume of gas V_m (L/mol) and $n = V / V_m$. - Ideal gas equation of state: $P \cdot V = n \cdot R \cdot T$.	I. Quantity of Matter (1. The mole unit, 2. Avogadro constant N _A , 3. Formula $n = N / N_A$) II. Molar Mass (1. Atomic molar mass, 2. Molecular molar mass, 3. Relation $n = m / M$) III. Gases & Ideal Gas Law (1. Molar volume V _m , 2. Avogadro-Ampere hypothesis, 3. Ideal gas law $P \cdot V = n \cdot R \cdot T$)	- Precision digital balance, iron nails, sulfur, table salt, glucose - Graduated syringes with pressure gauges, gas canisters - Graduated cylinders, beakers, periodic table	Activities: Weigh 1 mole of iron and 1 mole of water. Measure gas volume injected into syringe to verify $P \cdot V = n \cdot R \cdot T$. Evaluation: Calculations of moles from mass, volume, and pressure; Exam 4.

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Semester 2: Chemistry • Module 3: Solutions and Chemical Reactions (12 Hours)

Unit 8: Molar Concentration		4 Hours	
Prerequisites: The mole, molar mass, dissolution, mass concentration.		Problem Situation: Seawater and saline drips have precise salt concentrations. How do chemists prepare solutions of exact concentration by dissolution and dilution?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Define molar concentration of solute: $C = n / V$ (mol/L). - Relation with mass concentration: $C_m = C \cdot M$. - Protocol for preparing a solution by dissolution ($m = C \cdot V \cdot M$). - Protocol for dilution: conservation of matter $C_i \cdot V_i = C_f \cdot V_f$. - Dilution factor: $F = C_i / C_f = V_f / V_i$.	I. Molar Concentration (1. Definition, 2. Mass vs molar concentration) II. Preparation by Dissolution (1. Mass calculation, 2. Experimental glassware protocol) III. Dilution of Solutions (1. Conservation of quantity of matter, 2. Dilution factor F, 3. Glassware handling)	- Volumetric flasks (50, 100, 250 mL), volumetric pipettes (5, 10 mL) - Electronic balance, weighing boat, spatula, wash bottle - Potassium permanganate $KMnO_4$, copper sulfate $CuSO_4$	Activities: Calculate mass $m = C \cdot V \cdot M$, weigh solute, dissolve in volumetric flask, and dilute stock solution by factor $F = 10$. Evaluation: Dissolution and dilution exercises, glassware identification tests.

Unit 9: Chemical Transformations and Matter Balance		8 Hours	
Prerequisites: Chemical reaction, balancing equations, the mole, molar concentration.		Problem Situation: During combustion, reactants disappear while new products form. How do we track the exact quantity of each species and determine which reactant runs out first?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Model a chemical transformation with a balanced equation. - Define reaction progress x (in mol). - Construct an ICE reaction progress table. - Identify limiting reactant and maximum progress x_{max}. - Establish matter balance in final state.	I. Chemical Transformation (1. Initial and final state, 2. Reaction equation) II. Progress of Reaction (1. Definition of progress x , 2. ICE progress table) III. Matter Balance & Limiting Reactant (1. Hypotheses for x_{max} , 2. Limiting reactant determination, 3. Final composition)	- Reaction of magnesium ribbon with hydrochloric acid - Gas collection apparatus over water with graduated test tube - Precipitation reaction: copper sulfate + sodium hydroxide	Activities: React varied masses of Mg with fixed HCl volume; measure H_2 gas produced; verify that x_{max} matches theoretical limiting reactant calculation. Evaluation: Full ICE table problem solving and end-of-year Supervised Exam 6.